(3) H_n: $V = \mathbb{C}^n$ with basis v_i, $W = \mathbb{C}^{n-1}$ with basis w_i, $Av_i = w_i$, $Bw_i = v_{i+1}$ for $i < n$, and $Av_n = 0$.

(4) K_n is obtained from H_n by exchanging V with W and A with B.

Show that these are indecomposable and pairwise nonisomorphic.

(b) Show that if E is a representation of Q_2 such that AB is not nilpotent, then $E = E' \oplus E''$, where $E'' = E_{n,\lambda}$ for some $\lambda \neq 0$.

(c) Consider the case when AB is nilpotent, and consider the operator X on $V \oplus W$ given by $X(v, w) = (Bw, Av)$. Show that X is nilpotent and admits a basis consisting of chains (i.e., sequences $u, Xu, X^2u, \dots X^{l-1}u$ where $X^l u = 0$) which are compatible with the direct sum decomposition (i.e., for every chain $u \in V$ or $u \in W$). Deduce that (1)—(4) are the only indecomposable representations of Q_2.

(d) (Harder!) Generalize this classification to the Kronecker quiver, which has two vertices 1 and 2 and two edges both going from 1 to 2.

(e) (Still harder!) Can you generalize this classification to Q_n, $n > 2$ with any orientation? (Easier version: consider only the cyclic orientation).

Problem 6.9.2. Let $L \subset \frac{1}{2}\mathbb{Z}^8$ be the lattice of vectors where the coordinates are either all integers or all half-integers (but not integers) and the sum of all coordinates is an even integer.

(a) Let $\alpha_i = e_i - e_{i+1}$, $i = 1, \dots, 6$, $\alpha_7 = e_6 + e_7$, $\alpha_8 = -1/2 \sum_{i=1}^{8} e_i$. Show that α_i are a basis of L (over $\mathbb{Z}$).

(b) Show that roots in L (under the usual inner product) form a root system of type E_8 (compute the inner products of α_i).

(c) Show that the E_7 and E_6 lattices can be obtained as the sets of vectors in the E_8 lattice L where the first two, respectively three, coordinates (in the basis e_i) are equal.

(d) Show that E_6, E_7, E_8 have 72, 126, and 240 roots, respectively (enumerate types of roots in terms of the presentations in the basis e_i, and count the roots of each type).

Problem 6.9.3. Let V_α be the indecomposable representation of a Dynkin quiver Q which corresponds to a positive root α. For instance,